
***Course Name:** INT301: Operating Systems Fundamentals
– Week 2 Labs*

***Date of Submission:** 20/12/2025*

Part 1: Basic Scripting Exercises

Exercise 2.1: User Input Script

Task: Create a script that asks for your favorite color and food, then prints a sentence.

SCRIPT CODE :

```
echo "BY: LA'DESHA MITCHELLE"
```

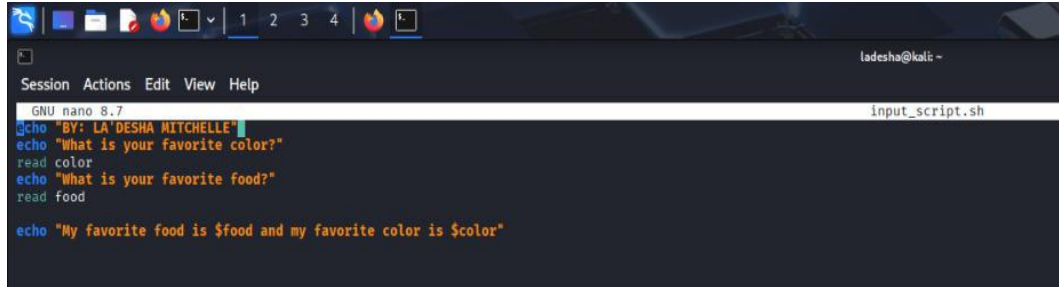
```
echo "What is your favorite color?"
```

```
read color
```

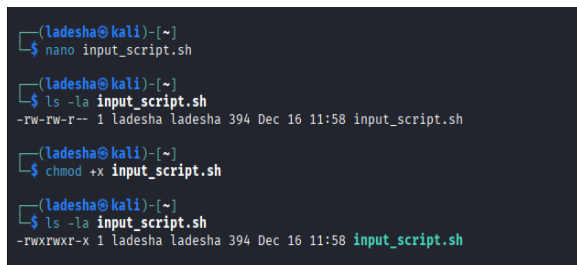
```
echo "What is your favorite food?"
```

```
read food
```

```
echo "My favorite food is $food and my favorite color is $color"
```

A screenshot of a terminal window with a dark background. At the top, there's a window title bar with icons and tabs labeled 1, 2, 3, 4. Below that is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The main area shows the GNU nano 8.7 editor editing a file named 'input_script.sh'. The code inside the editor is:

```
echo "BY: LA'DESHA MITCHELLE"
echo "What is your favorite color?"
read color
echo "What is your favorite food?"
read food
echo "My favorite food is $food and my favorite color is $color"
```

A screenshot of a terminal window showing a series of commands and their outputs. The commands are:

```
(ladesha@kali)-[~]
$ nano input_script.sh
(ladesha@kali)-[~]
$ ls -la input_script.sh
-rw-rw-r-- 1 ladesha ladesha 394 Dec 16 11:58 input_script.sh
(ladesha@kali)-[~]
$ chmod +x input_script.sh
(ladesha@kali)-[~]
$ ls -la input_script.sh
-rwxrwxr-x 1 ladesha ladesha 394 Dec 16 11:58 input_script.sh
```

A screenshot of a terminal window showing the output of the script execution. The commands and output are:

```
(ladesha@kali)-[~]
$ ./input_script.sh
BY: LA'DESHA MITCHELLE
What is your favorite color?
Black
What is your favorite food?
Pilau
My favorite food is Pilau and my favorite color is Black
```

Exercise 3.1: Password Checker

Task: Create a password checker for "secret123".

SCRIPT CODE :

```
echo "BY: LA'DESHA MITCHELLE"
```

```
echo "Enter your password:"
```

```
read password
```

```
if [ $password = "secret123" ]; then
```

```
    echo "Access granted"
```

```
else
```

```
    echo "Access denied"
```

```
fi
```

```

GNU nano 8.7 password_checker.sh
echo "BY: LA'DESHA MITCHELLE"
echo "Enter your password:"
read password

if [ $password = "secret123" ]; then
    echo "Access granted"
else
    echo "Access denied"
fi

```

```

(ladesha@kali)-[~]
└─$ nano password_checker.sh

(ladesha@kali)-[~]
└─$ ls -la password_checker.sh
-rw-rw-r-- 1 ladesha ladesha 140 Dec 16 13:18 password_checker.sh

(ladesha@kali)-[~]
└─$ chmod +x password_checker.sh

(ladesha@kali)-[~]
└─$ ls -la password_checker.sh
-rwxrwxr-x 1 ladesha ladesha 140 Dec 16 13:18 password_checker.sh

```

```

(ladesha@kali)-[~]
└─$ ./password_checker.sh
BY: LA'DESHA MITCHELLE
Enter your password:
37489
Access denied

(ladesha@kali)-[~]
└─$ ./password_checker.sh
BY: LA'DESHA MITCHELLE
Enter your password:
secret123
Access granted

```

Exercise 4.1: Countdown Script

Task: Count from 10 to 1, then print "Blast off!".

SCRIPT CODE :

```

echo "BY: LA'DESHA MITCHELLE"
count=10
message="Blast off!"
while [ $count -ge 1 ]
do
    echo "Count: $count"
    if [ $count -eq 1 ]; then
        echo "$message"
    else
        echo " "
    fi
    count=$((count - 1))
done

```

```
GNU nano 8.7 countdown_script.sh *
echo "By: LA'DESHA MITCHELLE"
count=10
message="Blast off!"
while [ $count -ge 1 ]
do
    echo "Count: $count"
    if [ $count -eq 1 ]; then
        echo "$message"
    else
        echo " "
    fi
    count=$((count - 1 ))
done

(ladesha@kali)-[~]
$ nano countdown_script.sh

(ladesha@kali)-[~]
$ ls -la countdown_script.sh
-rw-rw-r-- 1 ladesha ladesha 218 Dec 16 12:01 countdown_script.sh

(ladesha@kali)-[~]
$ chmod +x countdown_script.sh

(ladesha@kali)-[~]
$ ls -la countdown_script.sh
-rwxrwxr-x 1 ladesha ladesha 218 Dec 16 12:01 countdown_script.sh

(ladesha@kali)-[~]
$ ./countdown_script.sh
By: LA'DESHA MITCHELLE
Count: 10
Count: 9
Count: 8
Count: 7
Count: 6
Count: 5
Count: 4
Count: 3
Count: 2
Count: 1
Blast off!
```

Exercise 5.1: Sum Function

Task: Create a function that takes two numbers and returns their sum.

SCRIPT CODE :

```
echo "BY: LA'DESHA MITCHELLE"
calculate_sum() {
    local num1=$1
    local num2=$2
    local sum=$((num1 + num2))
    echo $sum
}
result=$(calculate_sum 5 8)
echo "The sum is: $result"
result=$(calculate_sum 7 3)
echo "The result is $result"
```

```
GNU nano 8.7 sum_function.sh *
echo "BY: LA'DESHA MITCHELLE"
calculate_sum() {
    local num1=$1
    local num2=$2
    local sum=$((num1 + num2))
    echo $sum
}

result=$(calculate_sum 5 8)
echo "The sum is: $result"

result=$(calculate_sum 7 3)
echo "The result is $result"
```

```
(ladesha@kali)-[~]
$ nano sum_function.sh

(ladesha@kali)-[~]
$ ls -la sum_function.sh
-rw-rw-r-- 1 ladesha ladesha 216 Dec 16 13:14 sum_function.sh

(ladesha@kali)-[~]
$ chmod +x sum_function.sh

(ladesha@kali)-[~]
$ ls -la sum_function.sh
-rwxrwxr-x 1 ladesha ladesha 216 Dec 16 13:14 sum_function.sh
```

```
(ladesha@kali)-[~]
$ ./sum_function.sh
BY: LA'DESHA MITCHELLE
The sum is: 13
The result is 10
```

Part 2: Practice Exercises

EASY EXERCISES

Greeting Script: Screenshot showing a time-based greeting.

SCRIPT CODE :

```
echo "BY: LA'DESHA MITCHELLE"

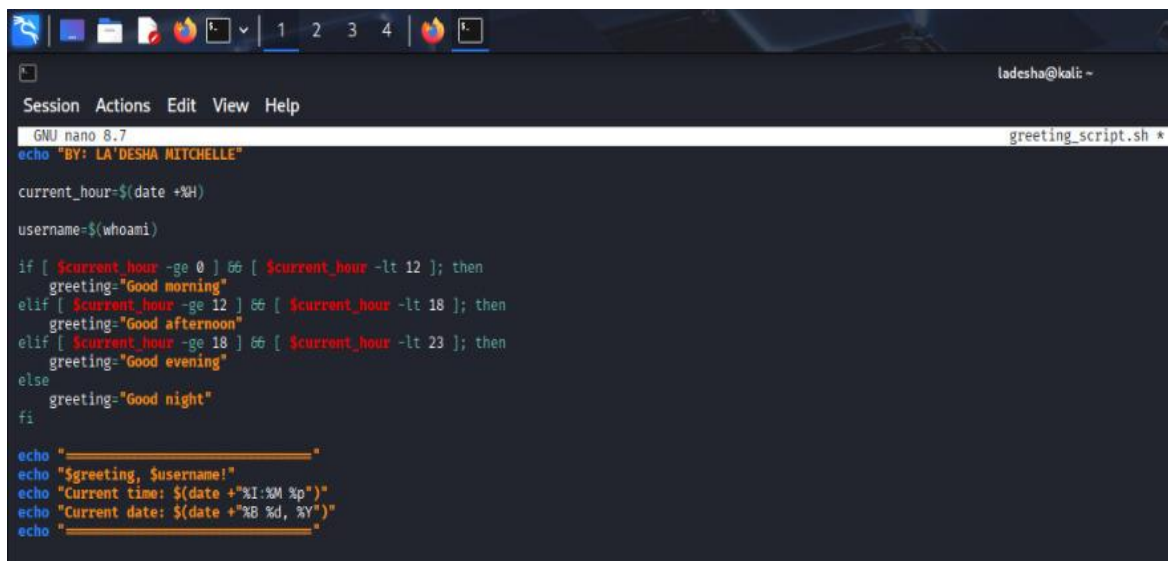
current_hour=$(date +%H)
username=$(whoami)

if [ $current_hour -ge 0 ] && [ $current_hour -lt 12 ]; then
    greeting="Good morning"
elif [ $current_hour -ge 12 ] && [ $current_hour -lt 18 ]; then
    greeting="Good afternoon"
```

```

elif [ $current_hour -ge 18 ] && [ $current_hour -lt 23 ]; then
    greeting="Good evening"
else
    greeting="Good night"
fi
echo "=====
echo "$greeting, $username!"
echo "Current time: $(date +%I:%M %p)"
echo "Current date: $(date +%B %d, %Y)"
echo "=====

```



```

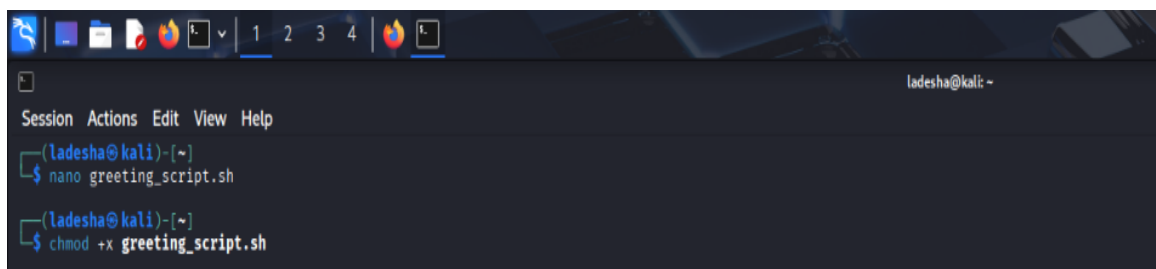
GNU nano 8.7 greeting_script.sh *
echo "BY: LA'DESHA MITCHELLE"

current_hour=$(date +%H)
username=$(whoami)

if [ $current_hour -ge 0 ] && [ $current_hour -lt 12 ]; then
    greeting="Good morning"
elif [ $current_hour -ge 12 ] && [ $current_hour -lt 18 ]; then
    greeting="Good afternoon"
elif [ $current_hour -ge 18 ] && [ $current_hour -lt 23 ]; then
    greeting="Good evening"
else
    greeting="Good night"
fi

echo "=====
echo "$greeting, $username!"
echo "Current time: $(date +%I:%M %p)"
echo "Current date: $(date +%B %d, %Y)"
echo "=====

```



```

(ladesha@kali)~$ nano greeting_script.sh

(ladesha@kali)~$ chmod +x greeting_script.sh

```



```

(ladesha@kali)~$ ./greeting_script.sh
BY: LA'DESHA MITCHELLE

=====
Good afternoon, ladesha!
Current time: 03:00 pm
Current date: December 17, 2025
=====

```

Number Guesser: Screenshot showing game interaction.

SCRIPT CODE :

```
#!/bin/bash

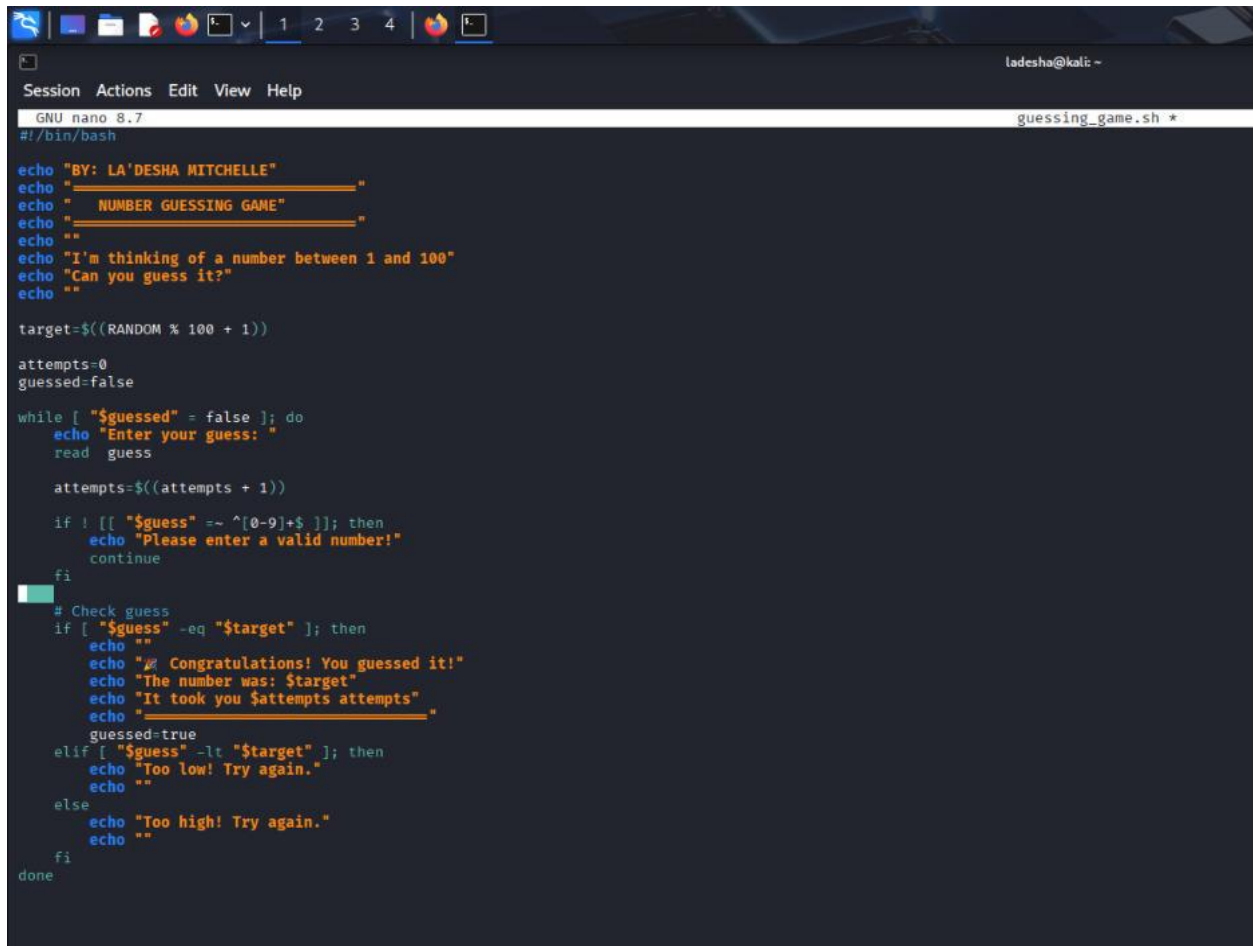
echo "BY: LA'DESHA MITCHELLE"
echo "===== "
echo "  NUMBER GUESSING GAME"
echo "===== "
echo " "
echo "I'm thinking of a number between 1 and 100"
echo "Can you guess it?"
echo " "

target=$((RANDOM % 100 + 1))
attempts=0
guessed=false

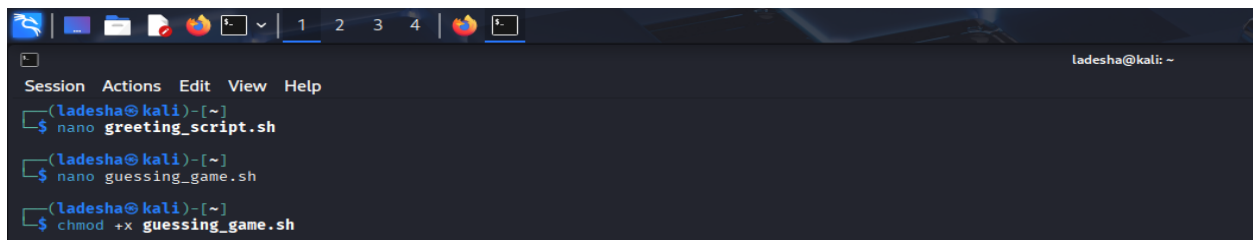
while [ "$guessed" = false ]; do
    echo "Enter your guess:"
    read guess

    attempts=$((attempts + 1))
    if ! [[ "$guess" =~ ^[0-9]+$ ]]; then
        echo "Please enter a valid number!"
        continue
    fi
    if [ "$guess" -eq "$target" ]; then
        echo " "
        echo " 🎉 Congratulations! You guessed it!"
        echo "The number was: $target"
        echo "===== "
        guessed=true
    elif [ "$guess" -lt "$target" ]; then
        echo "Too low! Try again."
        echo " "
    else
        echo "Too high! Try again."
```

```
echo ""  
  
fi  
done
```



```
GNU nano 8.7 guessing_game.sh *  
#!/bin/bash  
  
echo "BY: LA'DESHA MITCHELLE"  
echo "  
echo "    NUMBER GUESSING GAME"  
echo "  
echo "  
echo "I'm thinking of a number between 1 and 100"  
echo "Can you guess it?"  
echo "  
  
target=$((RANDOM % 100 + 1))  
  
attempts=0  
guessed=false  
  
while [ "$guessed" = false ]; do  
    echo "Enter your guess: "  
    read guess  
  
    attempts=$((attempts + 1))  
  
    if ! [[ "$guess" =~ ^[0-9]+$ ]]; then  
        echo "Please enter a valid number!"  
        continue  
    fi  
  
    # Check guess  
    if [ "$guess" -eq "$target" ]; then  
        echo "  
        echo "🎉 Congratulations! You guessed it!"  
        echo "The number was: $target"  
        echo "It took you $attempts attempts"  
        echo "  
        guessed=true  
    elif [ "$guess" -lt "$target" ]; then  
        echo "Too low! Try again."  
        echo "  
    else  
        echo "Too high! Try again."  
        echo "  
    fi  
done
```



```
ladesha@kali: ~  
$ nano greeting_script.sh  
  
ladesha@kali: ~  
$ nano guessing_game.sh  
  
ladesha@kali: ~  
$ chmod +x guessing_game.sh
```



```
(ladesha@kali)-[~]  
$ ./guessing_game.sh  
BY: LA'DESHA MITCHELLE  
  
=====
```

NUMBER GUESSING GAME

```
=====
```

I'm thinking of a number between 1 and 100
Can you guess it?

Enter your guess: 4
Too low! Try again.

Enter your guess: 23
Too low! Try again.

Enter your guess: 37
Too high! Try again.

Enter your guess: 29
Too low! Try again.

Enter your guess: 34
Too high! Try again.

Enter your guess: 31

🎉 Congratulations! You guessed it!
The number was: 31
It took you 6 attempts

```
=====
```

